# Yi Lin

## Software Engineer, Ph.D. in Computer Science

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Software engineer specializing in performance-critical systems, with expertise in language runtimes, garbage collection, and low-level optimization across the stack.

## Technical Skills

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|---|---|
| **Systems** | Language runtimes, virtual machines, garbage collection, compilers, memory management. |
| **Performance** | Profiling, benchmarking, low-level optimization, concurrency, performance analysis. |
| **Programming** | Rust, C/C++, Python, Go, TypeScript. |

## Employment

**2026 – 2020** | **Software Developer, Australian National University, Remote**

- Core team member of MMTk (`https://mmtk.io`), lead developer of `mmtk-core` (`https://github.com/mmtk/mmtk-core`).
  - Led the transformation of MMTk from a research prototype into a production-grade open-source project, enabling reusable high-performance garbage collection across diverse language runtimes, addressing a long-standing challenge in decoupling GC from language-specific runtime implementations.
  - Established rigorous engineering and evaluation workflows for MMTk projects, including cross-repository automated testing for performance and correctness, significantly improving reliability and development velocity.
  - Designed and evolved the architecture of `mmtk-core` to support diverse memory management requirements in high-performance applications and facilitate new GC algorithms.
  - Led performance profiling and root-cause analysis across the runtime stack (compiler, synchronization, and GC), identifying bottlenecks and implementing memory and time efficiency optimizations.
- Committer of Julia (`https://github.com/julialang/julia`).
  - Drove the integration of MMTk into the Julia runtime, bridging a production language runtime with a modular GC framework.
  - Partnered with JuliaHub to introduce MMTk-backed low-pause generational/concurrent GC into Julia.
  - Engineered key mechanisms to retrofit copying garbage collectors into Julia, unlocking support for collector designs widely adopted in high-performance language runtimes.
- My work enabled cross-language reuse of high-performance GC implementations, with adoption in production language runtimes (Ruby and Julia) as alternative collectors.

**2020 – 2018** | **Software Engineer, TracMap, Dunedin**

- Software Engineer at a GPS-guided agriculture technology company, building backend services for IoT fleet management and geospatial data processing.
  - Implemented microservice APIs in TypeScript and Go for device management and operational data processing.
  - Developed real-time spatiotemporal data processing pipelines for ingesting, transforming, and analyzing geospatial data streams.
  - Collaborated with product, design, and QA teams to deliver user-facing features with strong reliability and usability.
  - Supported AWS-based deployment and operations for scalable, resilient service infrastructure.

## Education

2018–2013 | **Ph.D. in Computer Science, Australian National University, Canberra**
- Thesis: An Efficient Implementation of a Micro Virtual Machine.
- Built a micro virtual machine (Zebu/Mu) achieving competitive performance relative to an LLVM-based backend, with 14.1% average overhead and most workloads within 1.5x, despite a significantly smaller (25K LOC) implementation.

2012–2011 | **M.Phil. in Computer Science, Australian National University, Canberra**

2011–2010 | **Master of Information Technology, University of Melbourne, Melbourne**

2009–2005 | **B.Eng. in Computer Science, Shanghai Jiao Tong University, Shanghai**

## Publications

2025 | **Reconsidering Garbage Collection in Julia, ISMM, Seoul**
- Luis E. de Souza Amorim, *Yi Lin*, et al.

2016 | **Rust as a Language for High Performance GC Implementation, ISMM, Santa Barbara**
- *Yi Lin*, Stephen M. Blackburn, et al.

2015 | **Stop and Go: Understanding Yieldpoint Behavior, ISMM, Portland**
- *Yi Lin*, Kunshan Wang, et al.

2015 | **Draining the Swamp: Micro Virtual Machines as a Solid Foundation for Language Development, SNAPL, Asilomar**
- Kunshan Wang, *Yi Lin*, et al.

2012 | **Unpicking the Knot: Teasing Apart VM/Application Interdependencies, VEE, London**
- *Yi Lin*, Stephen M. Blackburn, et al.